

1

Motion

In this chapter, we will learn about

- measurement of time and length
- expressing change in position in terms of distance and displacement
- distinguishing between scalars and vectors
- describing motion in terms of speed, velocity and acceleration
- uniform motion
- error estimation

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1.1 Time and length

1 Time

Flipped classroom
Watch a video and answer the questions.

Time has two meanings—the time at which an event occurs (time instant) and the time duration that an event lasts for (time interval). In this book, both time interval and time instant are represented by the letter *t*.

The *SI unit* of time is the **second**, written as **s**. Table 1.1a shows some other units of time. Digital stop-watches (Fig 1.1a) are commonly used to measure time intervals. The smallest time interval that they can measure is usually 0.01 s. Figure 1.1b shows different time intervals that we come across in daily life.

SI unit is the abbreviation of the International System of Units. It is the standard system of measurement in science. See Know more on p.5 for more details.



Fig 1.1a Digital stop-watch.

Name of unit	Symbol of unit	Conversion
hour	h	1 h = 3600 s
minute	min	1 min = 60 s
millisecond	ms	1 ms = 0.001 s = 10^{-3} s
microsecond	μs	1 μs = 0.000 001 s = 10^{-6} s

Table 1.1a Some units for time.

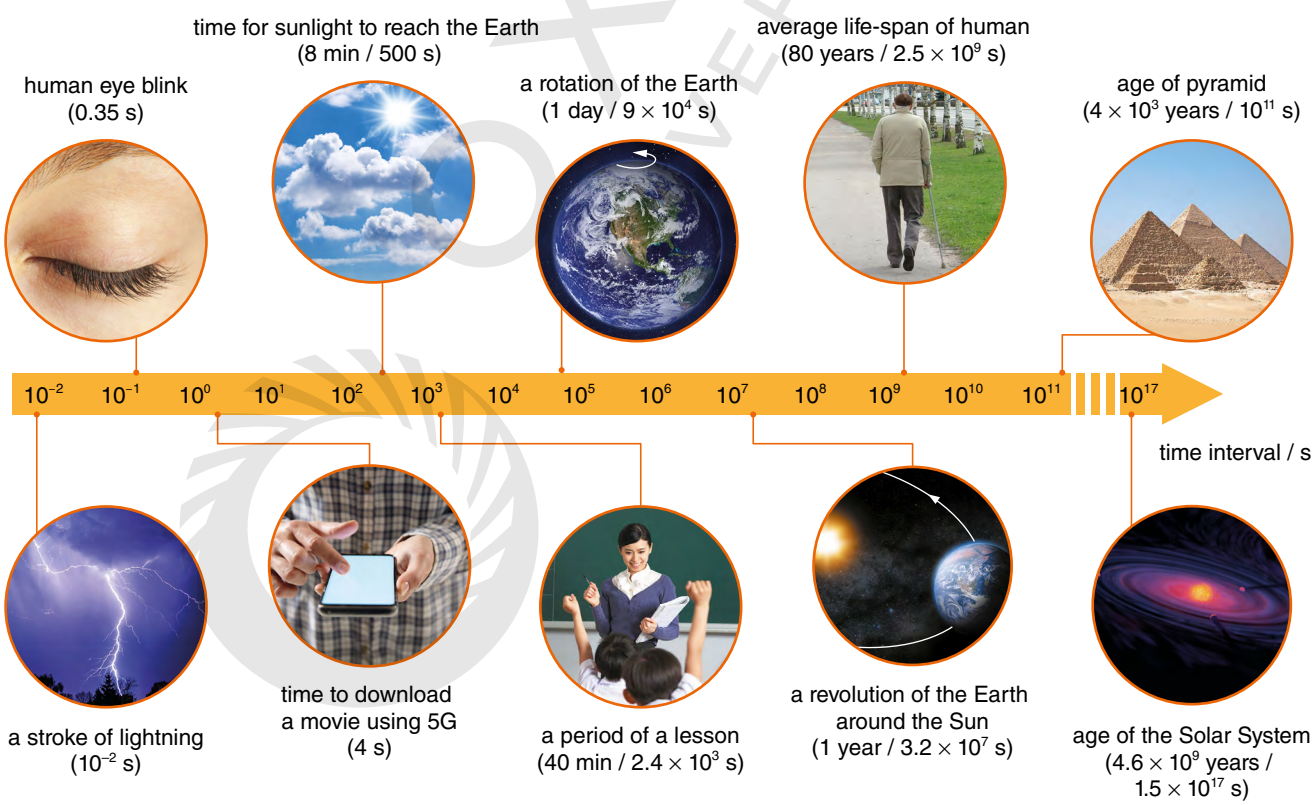


Fig 1.1b Some time intervals.

SI unit 國際單位制單位　second (s) 秒　millisecond (ms) 毫秒　microsecond (μs) 微秒

2 Length

Since the outbreak of COVID-19, people have had high awareness of the effectiveness of their masks. The BFE and PFE standards of masks (Fig 1.1c) are related to their abilities in blocking particles of sizes in the scale of 'micrometre'. How small is one micrometre?

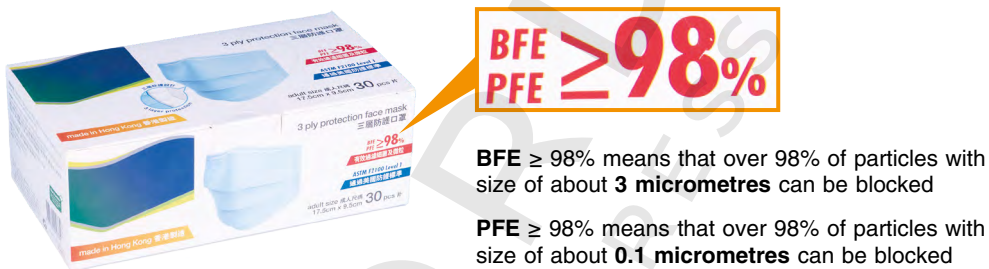


Fig 1.1c *A box of masks which achieves BFE and PFE standards.*

The metre is defined as the length of the path travelled by light in a vacuum during a time interval of $\frac{1}{299\ 792\ 458}$ of a second.

The SI unit of length is the **metre**, written as **m**. Table 1.1b shows some other *metric units* of length in different scales.

Name of unit	Symbol of unit	Conversion
kilometre	km	1 km = 1000 m = 10^3 m
centimetre	cm	1 cm = 0.01 m = 10^{-2} m
millimetre	mm	1 mm = 0.001 m = 10^{-3} m
micrometre	μm	1 μm = 0.000 001 m = 10^{-6} m
nanometre	nm	1 nm = 0.000 000 001 m = 10^{-9} m

Table 1.1b *Some metric units for different length scales.*

Refer to the question at the beginning. One micrometre is $\frac{1}{1000}$ of 1 mm.

Ts may tell Ss that extremely small and large lengths can be measured in other units (e.g. Å and light year) for convenience.
1 Å = 10^{-10} m
1 light year = 9.46×10^{15} m

Rulers or measuring tapes are commonly used to measure lengths (Fig 1.1d). Their finest gradation is usually 0.1 cm (i.e. 1 mm). This means that the measuring tools can measure the smallest length of 1 mm.

Figure 1.1e on p.4 gives some examples of a wide range of lengths.

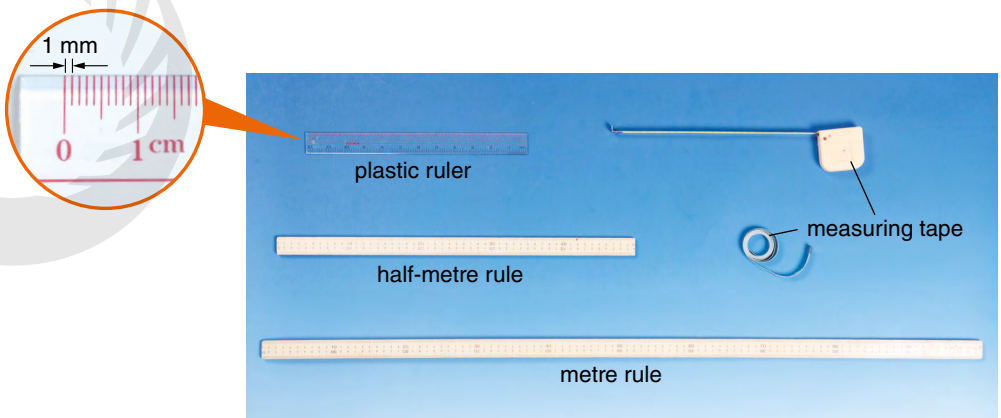


Fig 1.1d *Rulers and measuring tapes.*

metre (m) 米 metric unit 公制單位 micrometre (μm) 微米 nanometre (nm) 納米

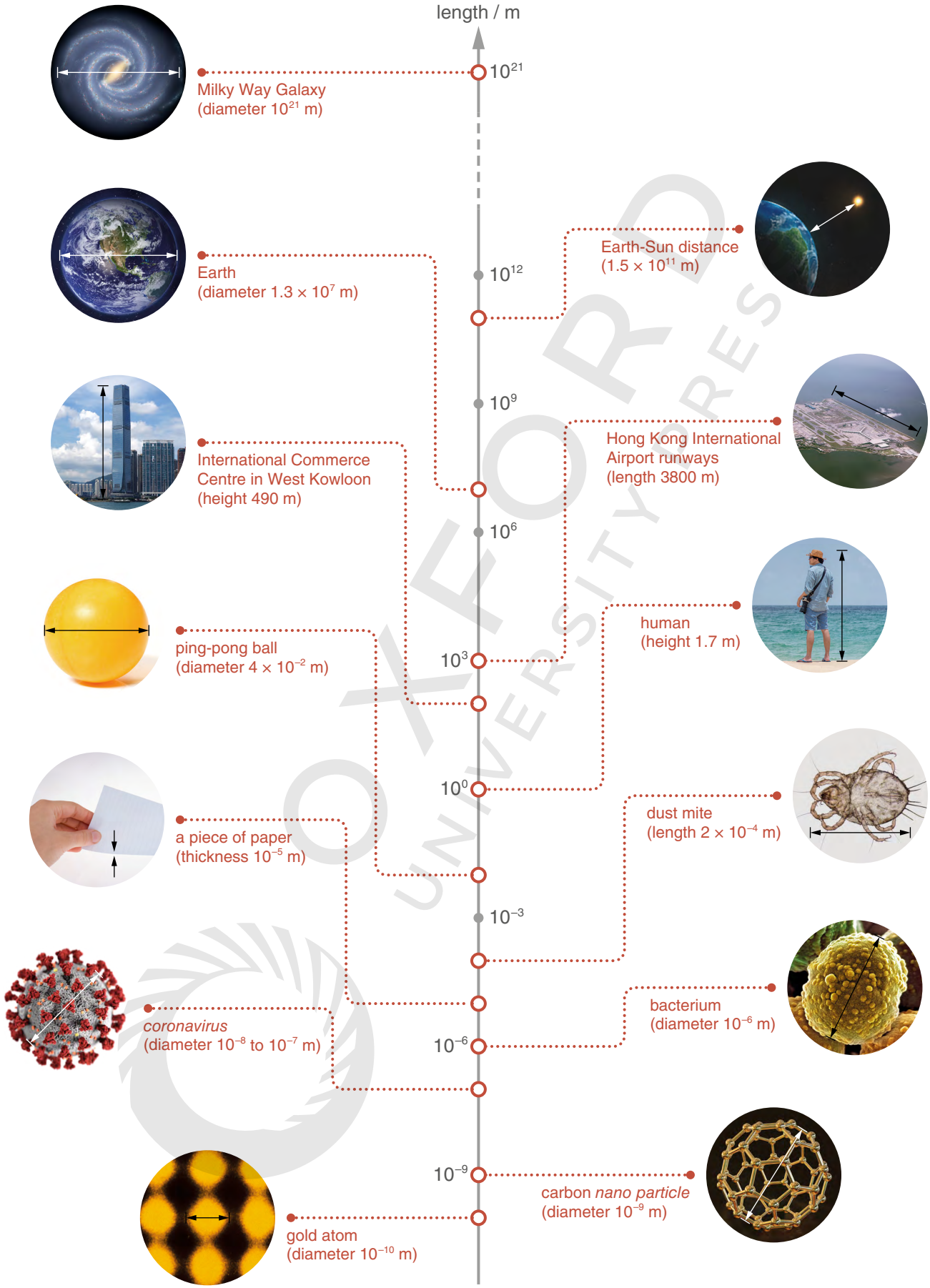


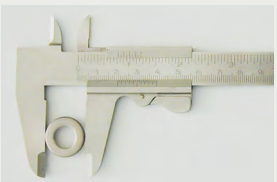
Fig 1.1e *Lengths in different scales.*

coronavirus 冠状病毒　　nano particle 纳米粒子

Know more

Other tools for measuring length

For more accurate measurements of length, a *vernier calliper* (accurate to 0.1 mm) or a *micrometer* (accurate to 0.01 mm) can be used.



A vernier calliper.



A micrometer.

Know more

Base SI units

Every day, we deal with lengths and time, which are usually expressed in metres (m) and seconds (s) respectively. These units are the base SI units of the standard metric system used in science. There are 7 base SI units (Fig a). All other units can be derived from the base units, e.g. the unit of density is kg m^{-3}.

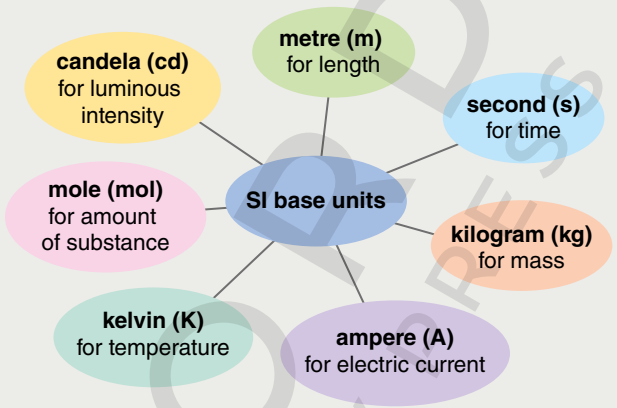


Fig a

Know more

Unit prefixes

For convenience, unit prefixes can be used to form units of various sizes. Some of them are listed below.

Prefix	nano	micro	milli	kilo
Symbol	n	μ	m	k
Meaning	10^{-9}	10^{-6}	10^{-3}	10^{3}

For example,

1 nm = 10^{-9} m = 0.000 000 001 m

1 μs = 10^{-6} s = 0.000 001 s

Exercise 1.1

1 State the SI unit of length and its symbol.

2 Convert the following lengths into the corresponding units.

(a) 3.2 mm = _____ m

(b) 1000 km = _____ m

(c) 0.1 m = _____ cm

(d) 2.85×10^{9} m = _____ km

3 State the SI unit of time and its symbol.

4 Convert the following times into the corresponding units.

(a) 51 μs = _____ s

(b) 432 min = _____ s

(c) 9000 s = _____ h

(d) 5×10^{-4} s = _____ μs

vernier calliper 游標尺　　micrometer 測微計

1.2 Distance and displacement

You may take many ways to go from point *X* in Tsim Sha Tsui to point *Y* in Central. Figure 1.2a shows some of the possible paths.

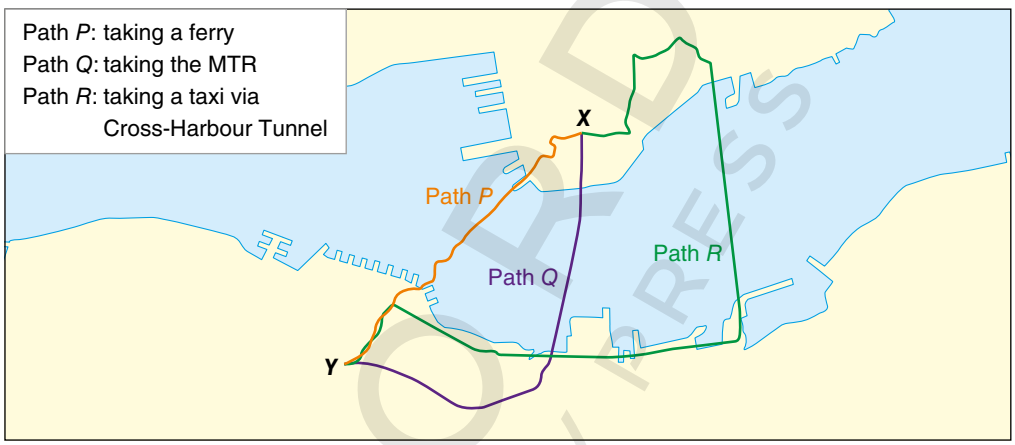


Fig 1.2a *Several paths from X to Y.*

Although the paths have different lengths, your overall changes in position are the same, i.e. from *X* to *Y*.

DSE goal

Describe the change of position in terms of distance and displacement.

Distinguish between scalar and vector.

Use scalars and vectors to represent physical quantities.

Simulation

→ This simulation shows the difference between distance and displacement. It shows the distance and displacement when a taxi moves. It is available on OUP web in PC version (.exe) only.

1 Displacement

In physics, when an object travels along a certain path,

the change in position of the object is called displacement.

On the other hand, the **distance** travelled by the object is the actual length of the path. Displacement can be represented by the letter *s* while distance can be represented by the letter *d*.

Suppose you jog from point *A* to *B*, then *C*, and then *D* (Fig 1.2b). The **distance travelled by you is equal to 225 m (= *AB* + *BC* + *CD*)**.

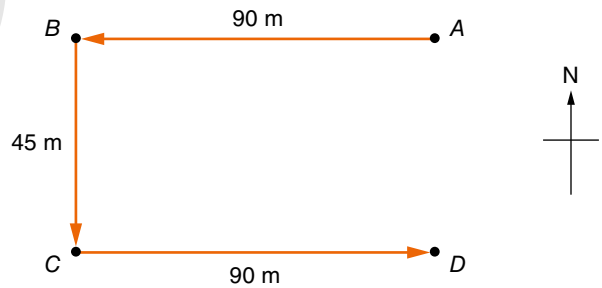


Fig 1.2b *Finding the distance travelled by adding the length of all sections (orange arrows) along the path.*

displacement 位移　　distance 距離

To find out your displacement, the following two things are required:

1 **Magnitude** (size): This is the length of the straight line joining the initial and final positions.

2 **Direction**: This is the direction of the straight line from the initial position to the final position.

Therefore, **your displacement (from *A* to *D*) is 45 m due south** (Fig 1.2c).

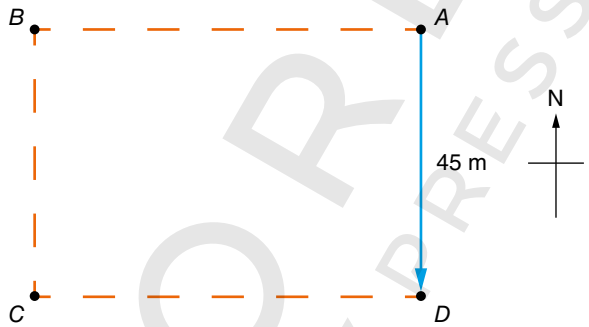


Fig 1.2c *Finding the displacement by considering the initial position (A) and the final position (D).*

Distance is a **scalar** quantity. It is described by magnitude only. By contrast, displacement is described by both magnitude and direction. This kind of quantity is called a **vector**.

A vector can be represented graphically by an arrow. In Figure 1.2c, the blue arrow represents the displacement vector from *A* to *D*. It can be written as $\overrightarrow{AD}$.

A scalar is described by magnitude only.
A vector is described by both magnitude and direction.

Distance tells us how long the path is but not the final position of the object (Fig 1.2d). Displacement depends solely on the initial and final positions. It tells us the exact position where the object moves to from an initial position, but we do not know which path the object takes (Fig 1.2e).

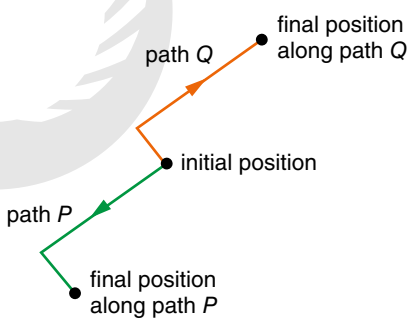


Fig 1.2d *Paths P and Q of the same length lead to different final positions.*

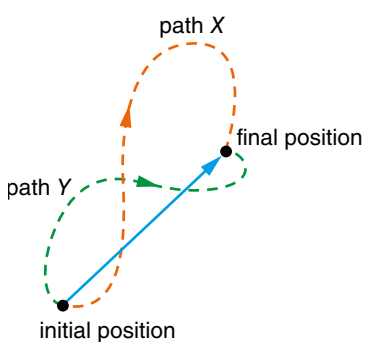


Fig 1.2e *Paths X and Y result in the same displacement.*

Know more

Distance posts

Distance posts can be found in some local nature trails. Do you know the meanings of the numbers on it?



▸ p.34

magnitude 量值 direction 方向 scalar 標量 / 無向量 vector 矢量 / 向量

2 Calculating displacement

To calculate the displacement of a journey, we need to find the magnitude and direction of the **vector pointing from the initial position to the final position**. The following shows the situations in one-dimension and two-dimension.

a Displacement along a straight line

Suppose you move along a straight line from B to A and then to C (Fig 1.2f). Your total displacement (or resultant displacement) from B is the vector pointing from B to C, i.e. $\overrightarrow{BC}$. It points towards the west and its magnitude BC is 40 m. Therefore, $\overrightarrow{BC}$ is 40 m (towards west).

$\overrightarrow{BC}$ denotes a vector while BC represents the magnitude of the vector. ▶

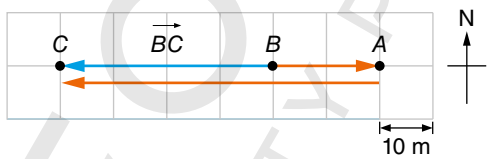


Fig 1.2f Calculating displacement along a straight line.

b Displacement on a plane

Suppose you move towards the north from D to E and then towards the east to F (Fig 1.2g). Your total displacement from D is $\overrightarrow{DF}$.

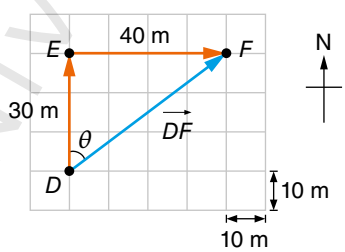


Fig 1.2g Calculating displacement on a plane.

The magnitude and direction can be found by using *Pythagoras' theorem* and *trigonometric ratios* as follows.

Magnitude:

$$DF = \sqrt{30^2 + 40^2} = 50 \text{ m}$$

Direction:

$$\tan\theta = \frac{40}{30}$$

$$\theta = 53.1°$$

As a result, $\overrightarrow{DF}$ is 50 m (N53.1°E).

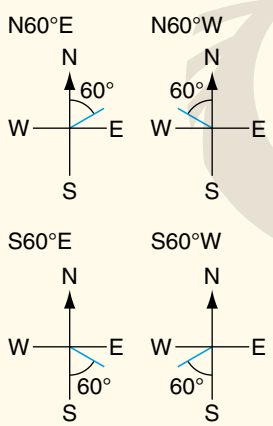
Skill *Mathematics*

Pythagoras' theorm and trigonometric functions

▶ p.451

Reduced bearings

Directions can be represented by *reduced bearings*. See the examples below.



→ Teaching notes and exercises are available on OUP web.

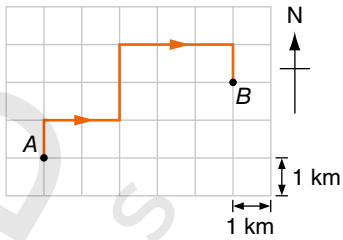
Pythagoras' theorem 畢氏定理 trigonometric ratio 三角比 reduced bearing 象限角

Example 1 Calculating displacement

A car travels from A to B along the path as shown.

(a) Find the total distance travelled by the car.

(b) Find the total displacement of the car.

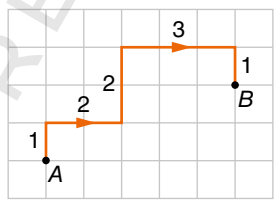


Solution

(a) Total distance travelled

$= 1 + 2 + 2 + 3 + 1$

$= 9$ km



(b) Magnitude:

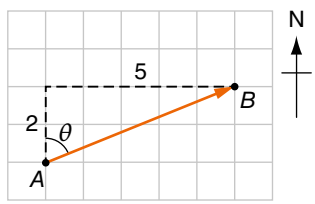
$AB = \sqrt{2^2 + 5^2} = 5.39$ km

Direction:

$\tan\theta = \frac{5}{2}$

$\theta = 68.2°$

The total displacement is 5.39 km (N68.2°E).



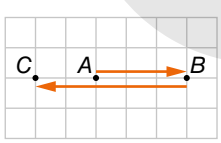
Remind Ss that the distance travelled will never be smaller than the magnitude of the displacement.

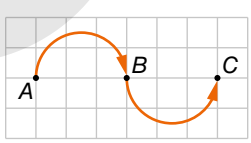
Think about...

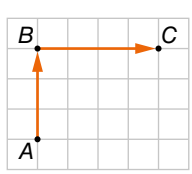
If another car travels from A to B such that its distance travelled is equal to the magnitude of its total displacement, what would its path be?

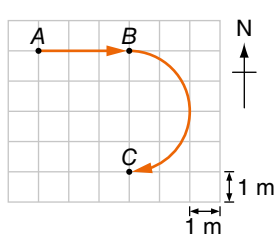
Learn by practice 1

1 In each of the following figures, an object moves from A to B then to C along the path as shown. In each case, find the **distance** travelled by the object and the **displacement** of the object. Given that the size of each grid is 1 m × 1 m.

(a) 

(b) 

(c) 

(d) 

Exercise 1.2

1 Which of the following statements about the distance travelled by an object and its displacement is correct?

A A longer distance travelled always means a displacement of a larger magnitude.

B A displacement of a larger magnitude always means a longer distance travelled.

C The magnitude of the displacement can be greater than the distance travelled.

D The distance travelled can be greater than the magnitude of the displacement.

***2** An ant moves from X to Y on the surface of a cube (Fig a). Each side of the cube is 10 cm long. Which of the following statements must be correct?

(1) The magnitude of the ant's displacement is 17.3 cm.

(2) The distance travelled by the ant is longer than 17.3 cm.

(3) The distance travelled by the ant is shorter than 30 cm.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

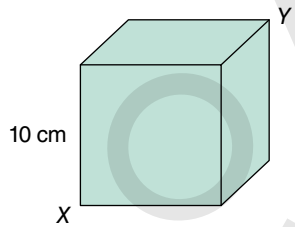


Fig a

3 Albert walks from X to Y along the orange path as shown (Fig b).

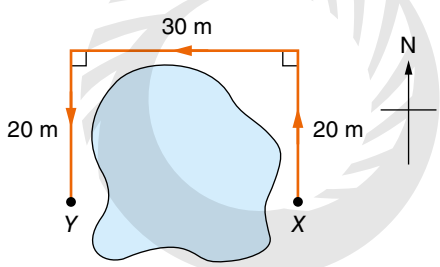


Fig b

(a) What is the distance that he travels?

(b) What is his displacement?

(c) Which of the above quantities is a scalar? Which one is a vector?

***4** Andy and Betty are playing Chinese chess. After several steps, Andy moves his horse from A to D (Fig c). Each square on the board is 2 cm in length.

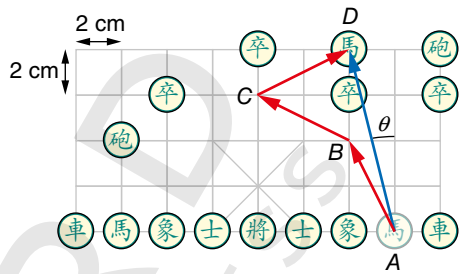


Fig c

(a) The total displacement of the horse is indicated by the blue arrow.

(i) Find its magnitude.

(ii) Find θ.

(b) Betty thinks that the displacement of Andy's horse is the same in each step. Is she correct? Explain your answer.

***5** Refer to Figure d.

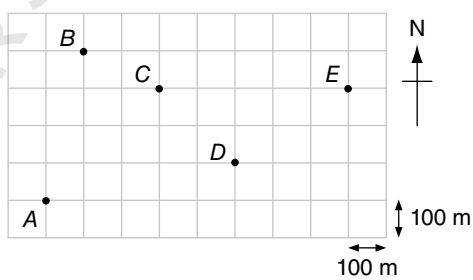


Fig d

(a) Find the displacement of the following objects.

(i) An object moves from A to E.

(ii) An object moves from C to A then to E.

(iii) An object moves from B to D then to C.

(iv) An object moves from D to E then to A.

(b) Can we find the distance travelled by the objects in (a)? Explain your answer.

1.3 Speed, velocity and acceleration

Suppose you want to measure your jogging speed with your smartphone. You may use an app that has the speedometer function (Fig 1.3a). You may also use an app that measures your total distance travelled and time spent. However, do the speeds obtained from these two apps have the same physical meaning?



speedometer　　recording distance and time

Fig 1.3a *Different smartphone apps for finding speed.*

DSE goal

Define average speed and average velocity.

Distinguish between instantaneous and average speed/velocity.

Describe motion of objects in terms of speed and velocity.

Solve problems involving displacement, time and velocity.

DSE exam

15(1A)Q4, 16(1A)Q4, 20(1A)Q4, 22(1A)Q4

1 Speed

Let us first recall what you learned in junior secondary science: **speed** is a measure of how fast an object moves.

$v = \frac{d}{t}$ ▶

Speed = distance travelled per unit time

Speed is a **scalar** and does not tell us the travelling direction of an object. The SI unit of speed is **metre per second**, written as **m s^{-1}**. Speed can be represented by the letter v.

The kilometre per hour (km h^{-1}) is another unit of speed.

1 m s^{-1} = 3.6 km h^{-1} ▶

$$1 \text{ km h}^{-1} = \frac{1 \text{ km}}{1 \text{ h}} = \frac{1000 \text{ m}}{60 \times 60 \text{ s}} = \frac{1}{3.6} \text{ m s}^{-1}$$

The speed limit on the local roads is 50 km h^{-1} unless traffic signs indicate otherwise. For example, the maximum speed allowed on the North Lantau Highway is 110 km h^{-1}. Figure 1.3b shows some speed limit signs.

✎ Ss may have a better feeling for the size of speed if it is expressed in km h^{-1}. They may know that the speed limits in Hong Kong are 50 km h^{-1} on ordinary roads, and 70–110 km h^{-1} on expressways.



Fig 1.3b *Speed limit signs.*

speed 速率

Figure 1.3c shows some typical speeds.

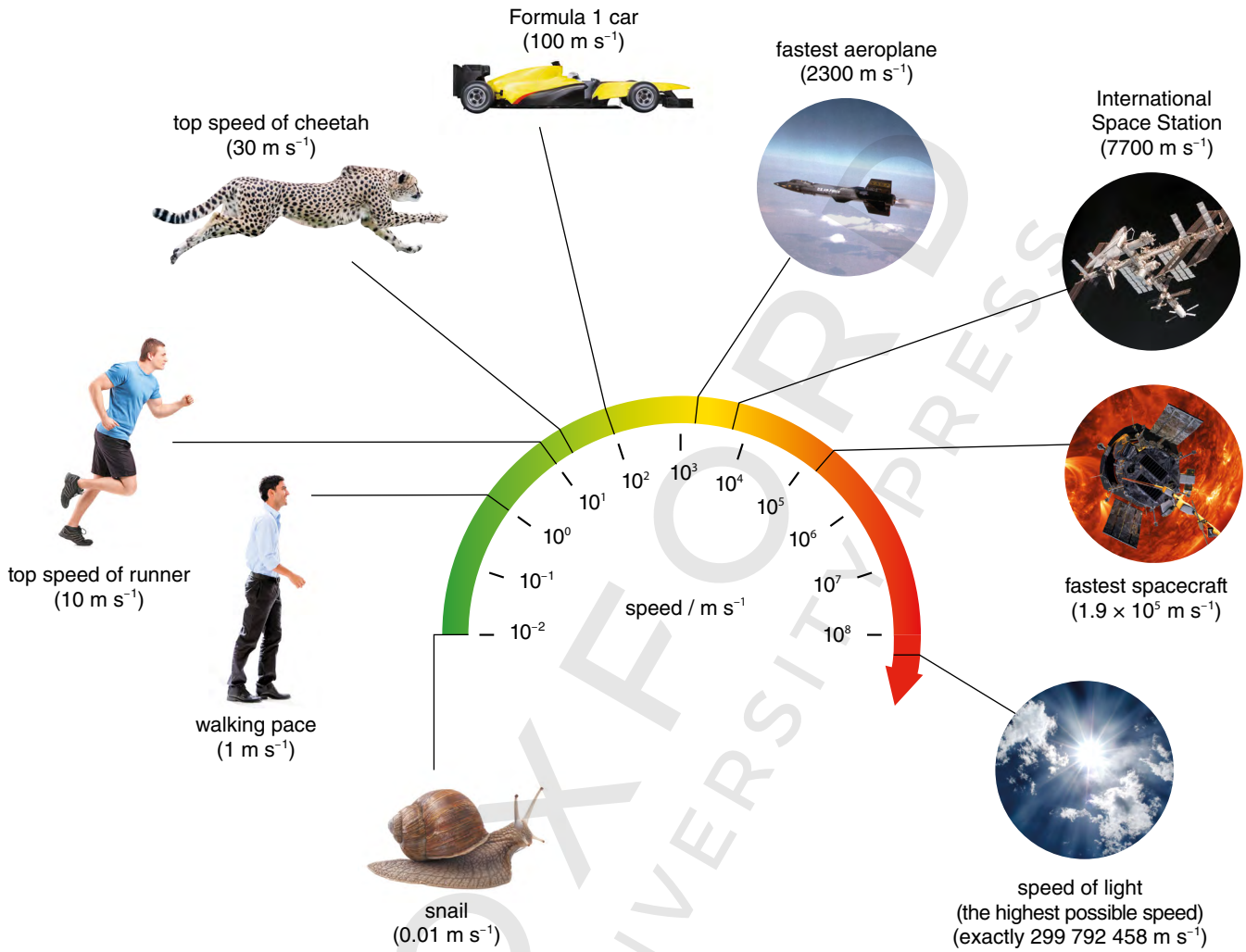


Fig 1.3c *Some typical speeds.*

a Instantaneous speed

The speed of an object may change. The speed **at an instant** is called the **instantaneous speed**. The speedometers in cars and the speed display units on minibuses show the instantaneous speeds (in km h^{-1}) of the vehicles (Fig 1.3d and 1.3e).

The speedometer in Fig 1.3a on p.11 also shows the instantaneous speed. ▶



Fig 1.3d *Speedometer in a car.*



Fig 1.3e *Speed display unit of a minibus.*

instantaneous speed 瞬時速率

→ This simulation shows the difference between instantaneous speed and average speed.

Simulation

b Average speed

Sometimes, we may want to know how fast an object moves **over a period of time**. In this case, we can find its **average speed**.

You can calculate your average speed using the second app in Fig 1.3a on p.11.

$v = \frac{d}{t}$

$$\text{Average speed} = \frac{\text{total distance travelled}}{\text{total time of travel}}$$

Example 2 | Average speed and speed limit

A car travels 7.2 km in 10 minutes.

(a) Find the average speed of the car. Give your answer in km h^{-1}.

(b) The speed limit on the road is 80 km h^{-1}. According to the answer in (a), can we conclude that the driver obeys the speed limit?

Solution

(a) Average speed $= \frac{d}{t}$

$$= \frac{7.2}{\frac{10}{60}}$$

$$= 43.2 \text{ km h}^{-1}$$

Tip
Time in min should be converted to time in h:

10 min $= \frac{10}{60}$ h

(b) No.

Tip
According to the information given, we are not sure whether the instantaneous speed of the car at any instant exceeds the speed limit.

If the instantaneous speed remains the same, it will be equal to the average speed. However, on most occasions, the instantaneous speed changes and is different from the average speed. Figure 1.3f further illustrates the difference between the two kinds of speeds.

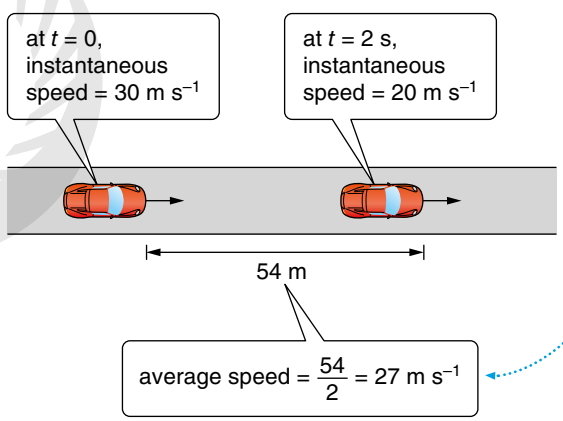


Fig 1.3f *Instantaneous speed and average speed.*

average speed 平均速率

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Determine whether the term 'speed' underlined in each of the following statements refers to an instantaneous speed or an average speed (Table a). Put a tick in the corresponding box.

	Instantaneous speed	Average speed
(a) Tom runs 100 m in 12 s. His speed is 8.33 m s^{-1}.		
(b) A car travels at a speed of 50 km h^{-1} at the moment t = 20 s.		
(c) The maximum road speed limit of a highway is 110 km h^{-1}.		

Table a

2 The table below shows the distance travelled L, time of travel t and average speed v of MTR trains along the Disneyland Resort Line and South Island Line. Complete the table.

	Disneyland Resort Line	South Island Line
L / km		7.4
t / s	360	720
v / m s^{-1}	9.72	
v / km h^{-1}		

Table b

3 The instantaneous speeds of an object at t = 1 s and t = 2 s are 5 m s^{-1} and 7 m s^{-1} respectively. What is its average speed from t = 1 s to t = 2 s?

A 4 m s^{-1} B 6 m s^{-1}

C 12 m s^{-1} D Cannot be determined

Remind Ss that speed and velocity carry the same meaning in daily language. However, in physics, they refer to different concepts: speed is a scalar and velocity is a vector.

$v = \frac{s}{t}$ ▶

2 Velocity

Velocity tells us how fast and in what direction the position of an object changes.

Velocity = displacement per unit time

Velocity is a **vector** and its SI unit is m s^{-1}. To express the velocity fully, we have to include both magnitude and direction. Velocity can also be represented by the letter v.

velocity 速度

a Instantaneous velocity

Like the instantaneous speed, the velocity at an instant is called the **instantaneous velocity**. Its magnitude is the same as the instantaneous speed and its direction is the direction of motion of the object at that instant (Fig 1.3g).

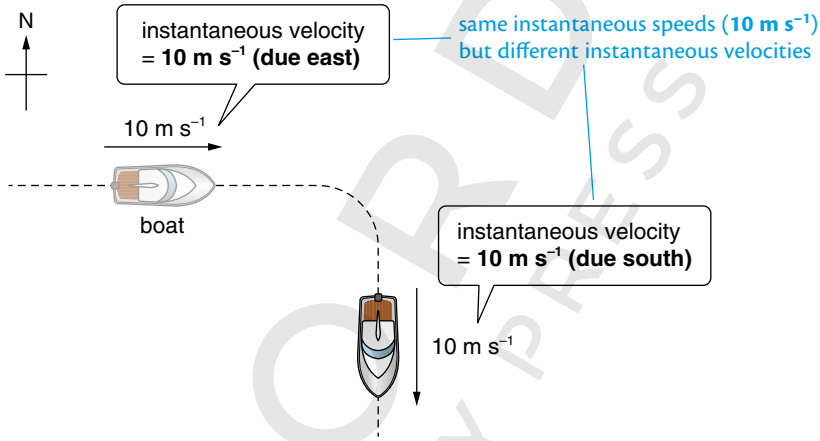


Fig 1.3g *Instantaneous velocity and instantaneous speed.*

b Average velocity

The **average velocity** tells us how fast and in what direction the position of an object changes over a period of time.

$v = \frac{s}{t}$ ▶

$$\text{Average velocity} = \frac{\text{total displacement}}{\text{total time of travel}}$$

The direction of the average velocity is always the same as that of the total displacement (Fig 1.3h).

Average speed ▶

$= \frac{100}{10} = 10 \text{ m s}^{-1}$

Magnitude of average velocity

$= \frac{80}{10} = 8 \text{ m s}^{-1}$

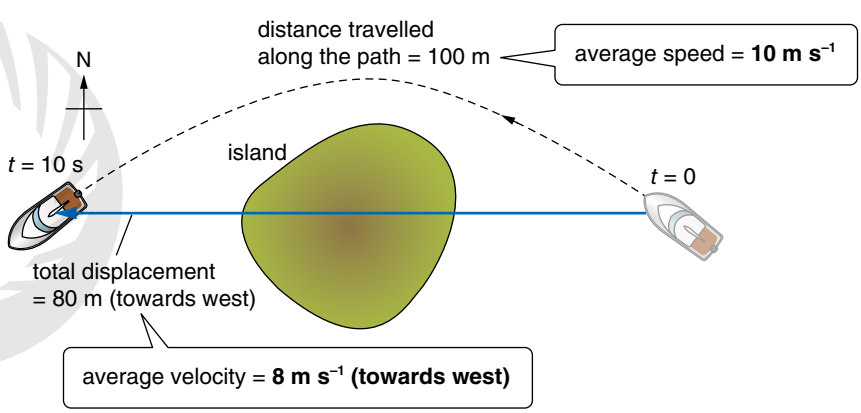


Fig 1.3h *Average velocity and average speed.*

Note that **the instantaneous velocity can be estimated by measuring the average velocity over an extremely short time interval.**

instantaneous velocity 瞬時速度　　average velocity 平均速度

→ Teaching notes and exercises are available on OUP web.

Skill
Problem-solving

Finding average velocity

Follow the steps below to find the average velocity.

① Draw the path of the object.

② Draw the total displacement (an arrow from the initial position to the final position).

③ Find the magnitude and direction of the average velocity.

Example 3 Average speed and average velocity

A bicycle travels 800 m north and then 600 m west in 400 s.

(a) Find its average speed.

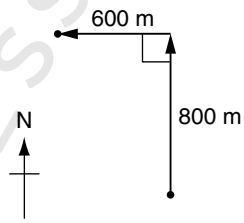
(b) Find its average velocity.

Solution

(a) Average speed $= \frac{d}{t}$

$= \frac{800 + 600}{400}$

$= 3.5 \text{ m s}^{-1}$



(b) Magnitude:

Magnitude of total displacement $= \sqrt{800^2 + 600^2} = 1000 \text{ m}$

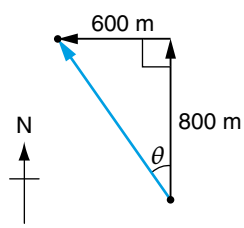
Magnitude of average velocity $= \frac{s}{t} = \frac{1000}{400} = 2.5 \text{ m s}^{-1}$

Direction:

$\tan \theta = \frac{600}{800}$

$\theta = 36.9°$

The average velocity is 2.5 m s^{-1} (N36.9°W).

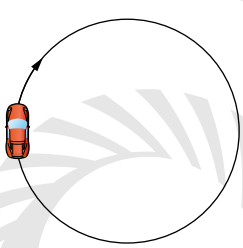


Quick check 2

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 A car keeps moving in a circle at 5 m s^{-1}.



Which of the following statements is/are correct? Put a tick in the corresponding box(es).

- ☐ Its instantaneous speed remains unchanged.
- ☐ Its instantaneous velocity remains unchanged.
- ☐ The magnitude of its instantaneous velocity remains unchanged.
- ☐ The direction of its instantaneous velocity remains unchanged.

2 Which of the following statements about a moving object is/are correct? Put a tick in the corresponding box(es).

- ☐ Its instantaneous speed can be greater than the magnitude of its instantaneous velocity at a certain instant.
- ☐ The magnitude of its average velocity can be greater than its average speed in a certain period of time.
- ☐ Its average velocity in a certain period of time can be larger than its instantaneous velocity at a certain instant within that period of time.
- ☐ Its average velocity and displacement within a certain period of time always point in the same direction.

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Learn by practice 2

1 An object travels a distance of 50 m in 10 s. Then it travels a distance of 30 m in 5 s. What is its average speed

(a) in the first 10 s,

(b) in the last 5 s,

(c) in the whole journey?

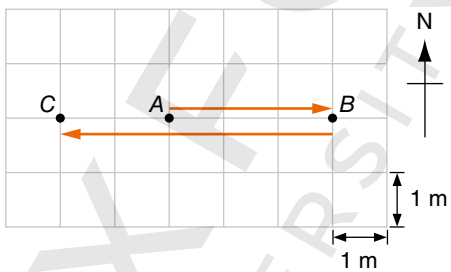
2 A taxi is travelling on a road which has a speed limit of 50 km h^{-1}. It travels 1.8 km in 2 minutes.

(a) Find the average speed of the taxi in km h^{-1}.

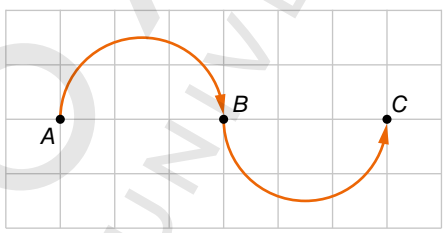
(b) Does the taxi violate the speed limit? Explain your answer.

3 In each of the following figures, an object takes 4 s to move from *A* to *B*, and then takes 2 s to move from *B* to *C* along the path as shown. In each case, find the **average speed** and **average velocity** of the object in its journey. Given that the size of each grid is 1 m × 1 m.

(a)

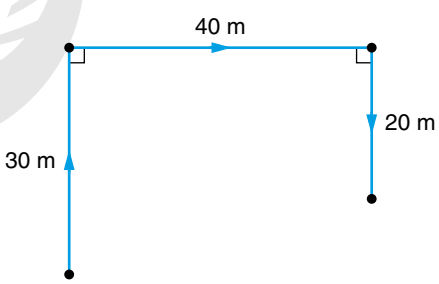


(b)

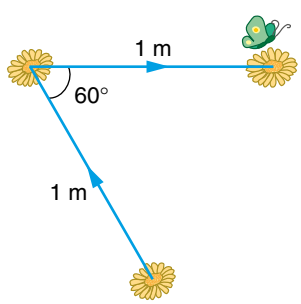


4 Find the **average speed** and the **magnitude of the average velocity** in each of the following cases.

(a) A remote-controlled car takes 30 s to travel along the path as shown.



(b) A butterfly flies along the path as shown in 20 s.



DSE goal
Define acceleration.

DSE exam
6(1A)Q5, 17(1A)Q5

3 Acceleration

Acceleration is defined as the rate of change of velocity. It is a measure of how fast an object changes its velocity.

Acceleration = change in velocity per unit time

Acceleration is a **vector** and its SI unit is **metre per second squared**, written as **m s^{-2}**. It is in the **same direction as the change in velocity**.

Although people usually refer to acceleration as 'speeding up' in daily conversation, the term carries a broader meaning in physics. An object accelerates **whenever its velocity changes** (Fig 1.3i), i.e. whenever it

1 speeds up;

2 slows down;

3 changes direction.

Sometimes we speak of *deceleration*. An object decelerates when it slows down.

✎ The term 'deceleration' (or 'decelerate') is often used in public exams (e.g. DSE 2016 Paper 1A Q7, DSE 2012 Paper 1B Q4). It is used as the magnitude of the acceleration of an object which is slowing down. In this sense, deceleration will not be negative.

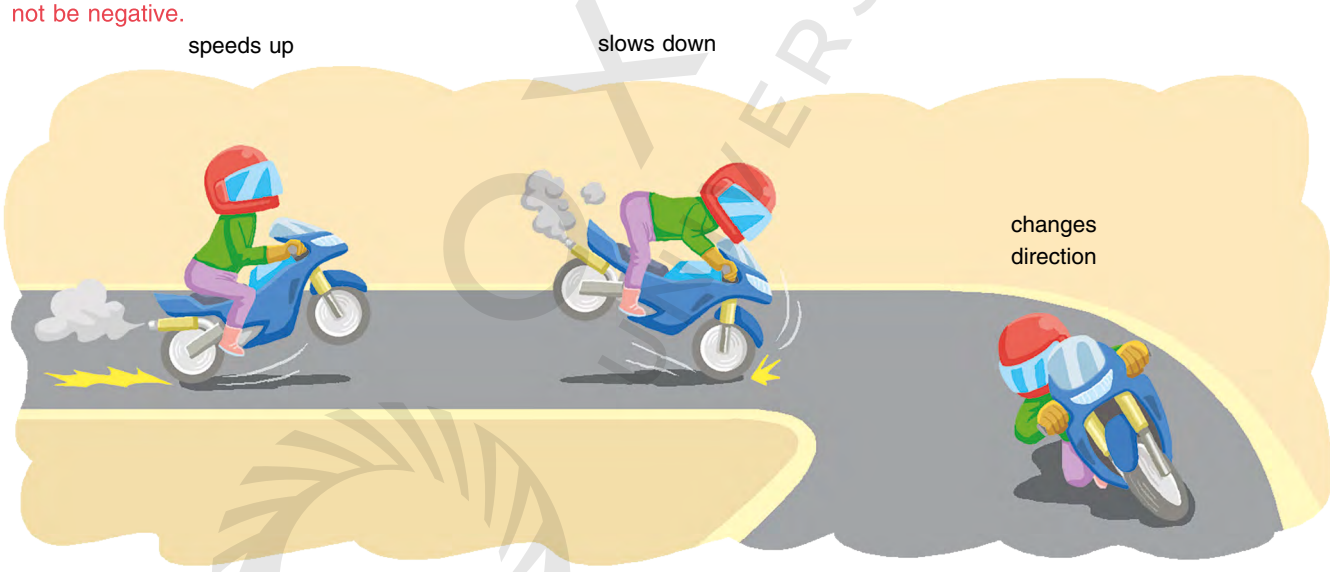


Fig 1.3i *A motorcycle accelerates when it speeds up, slows down or changes direction.*

acceleration 加速度　　deceleration 減速度

In Figure 1.3j, a taxi and a sports car **speed up from rest in the same direction** along a straight road. The taxi accelerates at 2 m s^{-2} and the sports car accelerates at 4 m s^{-2}.

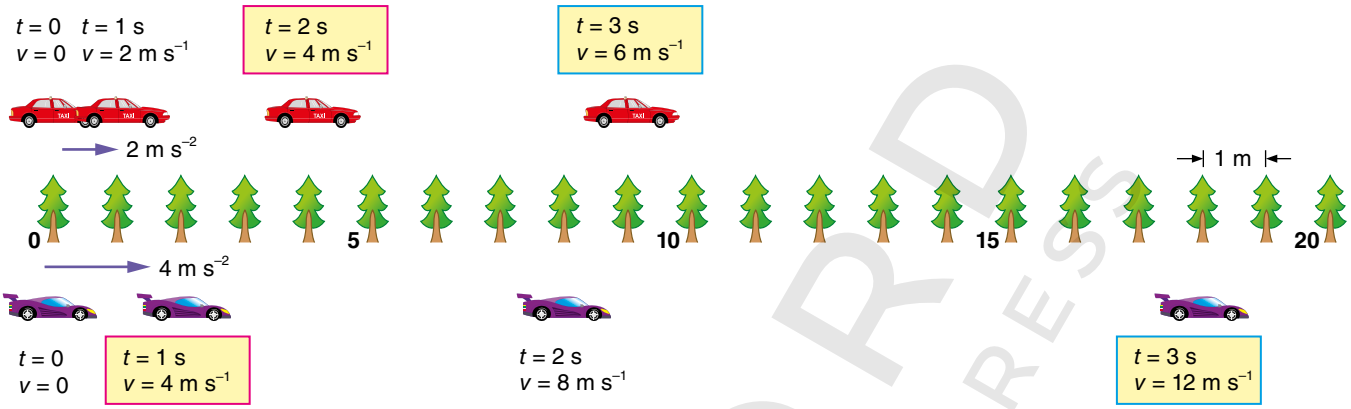


Fig 1.3j *Cars start from rest with different constant accelerations in the first 3 seconds (v represents speed).*

We can see that:

1. The speeds of the taxi and the sports car increase by 2 m s^{-1} and 4 m s^{-1} each second respectively.
2. The sports car takes less time to reach a certain speed, e.g. from v = 0 to v = 4 m s^{-1} (red boxes).
3. The distance travelled by each car per second increases with time.
4. The sports car travels a longer distance in the same period of time, e.g. from t = 0 to t = 3 s (blue boxes).

Figure 1.3k shows some typical magnitudes of acceleration.

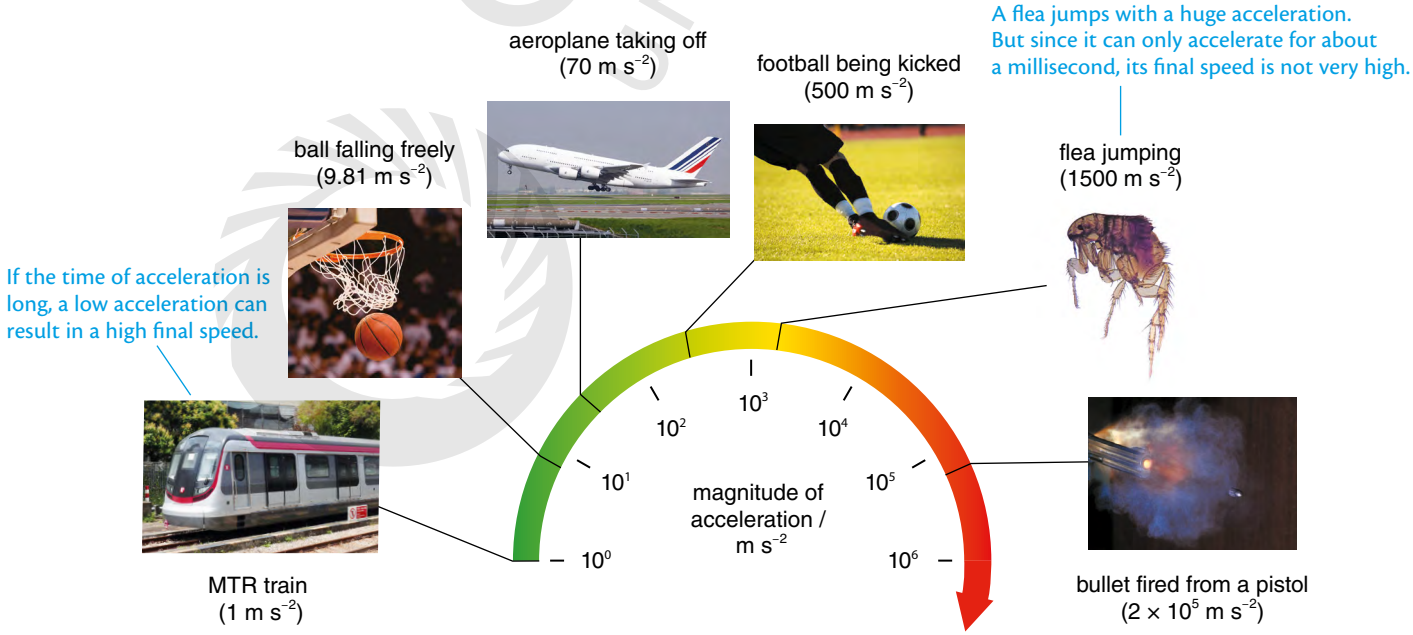


Fig 1.3k *Some typical magnitudes of acceleration.*

DSE goal

Interpret uniform motion of objects.

4 Uniform motion

When a skater glides on the ice at a **constant velocity**, she is said to be in **uniform motion** (Fig 1.3l). In this special case of motion, the acceleration is zero. Moreover, the instantaneous velocity equals the average velocity.



Fig 1.3l A skater in uniform motion.

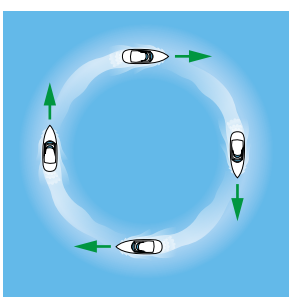


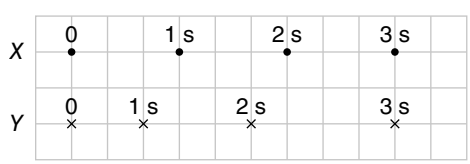
Fig 1.3m A boat moving along a circular path.

Note that an object moving at a constant speed is not necessarily in uniform motion. Figure 1.3m shows a boat moving along a circular path at a constant speed. In each position shown, the boat is moving in a different direction; in fact its velocity keeps changing. Therefore, it is not in uniform motion.

For uniform motion, velocity = constant (speed = constant; moving direction unchanged)
acceleration = 0

Example 4 Uniform motion

The positions of object X (represented by •) and Y (represented by ×) at time t = 0, 1 s, 2 s and 3 s are as shown.



One of the objects is in uniform motion. Which one is it? Explain briefly.

Tip

The distance travelled by X in each second remains the same and it keeps moving in the same direction.

On the other hand, the distance travelled by Y in each second increases, showing that it accelerates.

Solution

X is in uniform motion as its velocity is constant.

uniform motion 勻速運動

Quick check 3

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

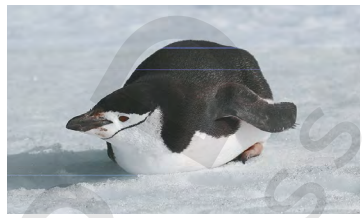
1 Which of the following statements about a moving object is/are correct? Put a tick in the corresponding box(es).

- ☐ The acceleration of an object travelling at a constant speed must be zero.
- ☐ If an object has a non-zero acceleration, its speed must be changing.
- ☐ An object must be in uniform motion if its speed is constant.
- ☐ The speed of an object must be constant if it is in uniform motion.

2 *True or false:* If object X has a larger magnitude of acceleration than object Y at a certain instant, X must have a higher speed than Y at that instant. (T/F)

3 Are the following animals in uniform motion?

(a) A penguin sliding at a constant speed along a straight line



(b) A polar bear at rest



Learn by practice 3

1 A girl skis along a straight path and speeds up with an acceleration of 0.5 m s^{-2}. Complete Table a to show the magnitude of her velocities at different time instants.

Time / s	0	1	2	3
Velocity / m s^{-1}	0		1	

Table a

2 A boy skis in uniform motion with a speed of 2.5 m s^{-1}. At time t = 0, he is at position X. Complete Table b to show the magnitude of his displacement from X at different time instants.

Time / s	0	2	12	40
Displacement / m	0			

Table b

Exercise 1.3

1 An object moves at a constant speed for a certain period of time. Which of the following quantities must have the same value?

(1) Its average speed

(2) The magnitude of its average velocity

(3) The magnitude of its instantaneous velocity

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

2 A girl sliding on a wet plastic sheet is in uniform motion (Fig a). It takes her 0.5 s to slide 1.5 m. How long does she take to slide 0.9 m?



Fig a

A 0.2 s B 0.3 s

C 0.6 s D 0.9 s

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3



Fig b

The acceleration of a cheetah (Fig b) is higher than that of a sports car. Which of the following can be concluded from this?

(1) A cheetah can reach a higher maximum speed.

(2) A cheetah can increase its speed by the same amount in a shorter time.

(3) A cheetah can move a longer distance within the same period of time.

A (1) only B (2) only

C (2) and (3) only D (1), (2) and (3)

4 An object is in uniform motion. Its position at $t = 0$ and 3 s are shown (Fig c). Draw its positions at $t = 1$ s and 2 s in the figure.



Fig c

5 Voyager 1 (Fig d) is the first space probe to travel beyond the solar system. It is now in uniform motion and travels about 1000 km forwards each minute.



Fig d

(a) Find the magnitude of its velocity.

(b) Find the magnitude of its displacement in 1 hour.

6 Susan runs at 6 m s^{-1} for 8 s, and then runs at 8 m s^{-1} for 2 s. Find her average speed over the whole journey.

7 Sarah Lee Wai Sze (Fig e) won the gold medal with a time of 33.755 s in a 500-metre time trial track cycling event. The track is 250 m long and she completed 2 laps in the event.



Fig e

(a) Find her average speed in (i) km h^{-1}, (ii) m s^{-1}.

(b) What is the magnitude of her average velocity in the whole competition?

★**8** A man walks at 2 m s^{-1} for 10 m from A to B, and then walks from B back to A along the same path at 1 m s^{-1}. Find the average speed of the man over the whole journey.

★**9** A squirrel moves at 0.2 m s^{-1} horizontally from A to B in 3 s (Fig f), and then moves at 0.25 m s^{-1} vertically upwards to C in 2 s.



Fig f

(a) Find the average speed of the squirrel.

(b) Find the magnitude of the average velocity of the squirrel.

1.4 Error estimation

In the school athletic meet, there are lots of student helpers in track races to measure the time of completion of different athletes. However, sometimes there may be strange results obtained, as illustrated in Figure 1.4a.

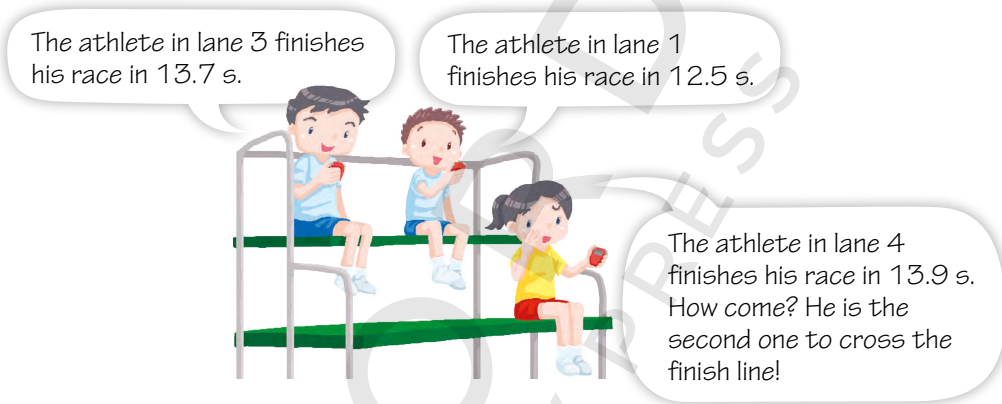


Fig 1.4a *Strange results obtained by student helpers in track races.*

No matter how carefully the measuring process is done, error always exists in the process and cannot be avoided. It is a part of the measurement. It is not the same as a mistake—which is avoidable.

The reason for this situation is the existence of *errors*. Errors exist in all measurements and can affect the result obtained. The following part mainly focuses on error estimation.

1 Reaction time

Before discussing the errors in taking measurements, let us first do the following activity.

Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

Grasp to win

What to do

1 Place an object on the table. Place one hand on the table close to the object, and your teacher would place his/her hand on the table further from the object.

2 It is a competition to see who can grasp the object first. Your teacher can start at any time. You can only start moving your hand when your teacher's hand has left the table.



What you find

After several trials, do you think it is easy for you to win this game?

error 誤差

Simulation
→ This simulation shows a series of tests that can obtain your average reaction time. It is available on OUP web in PC version (.exe) only.

In this activity, you probably find it difficult to win. This activity shows that there is a time lapse between seeing an event and taking action to respond. This time lapse, called *reaction time*, is about 0.2 s for a normal person and is the main source of error in using a digital stop-watch. This source of error does not come from the measuring tool but the person controlling the tool (Fig 1.4b).



Fig 1.4b *Different people may have different reaction times.*

To show how significant an error is in measurement, we can find the percentage error.

$$\text{Percentage error} = \frac{\text{error}}{\text{measured value}} \times 100\%$$

For repeating events with each event taking the same duration, the percentage error due to reaction time can be reduced by measuring the duration of more events. See Example 5 on p.25.

Know more

Automatic timing in sports

In 1968, electronic timing first became the primary timing method in the Olympic Games. However, the timing systems during that time had a delay and adjustments were needed. Nowadays, the electronic timing devices used in the Olympic Games have been greatly improved. The official results are rounded off to the nearest 0.01 s.

The high accuracy in timing can be achieved by different methods. In a sprint race, for example, the time is automatically started by the set-off signal. A camera (Fig a) aimed at the finish line takes images at extremely high frame rate (up to 10 000 frames each second). A timestamp is given to each image. The time for each athlete can be known after analysing the images (Fig b).



Fig a

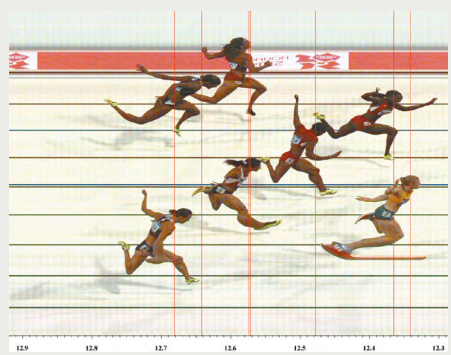


Fig b

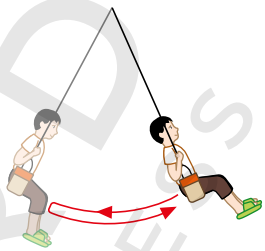
reaction time 反應時間

→ This is a Video Quiz about the experiment 'Reducing percentage error'.

Example 5 Percentage error in timing

Polly wants to use a stop-watch to find out the time T for a swing to move one cycle.





(a) Polly uses the stop-watch to measure the time for the swing to move one cycle directly. Her result is 2.31 s. If the error due to reaction time is 0.4 s, what is the percentage error due to reaction time?

(b) She then uses the stop-watch to measure the time for the swing to move 10 cycles and the result is 22.48 s. The erroe due to reaction time is still 0.4 s.

(i) What is the value of T according to this measurement?

(ii) What is the percentage error due to her reaction time in this measurement?

Solution

(a) Percentage error $= \frac{0.4}{2.31} \times 100\% = 17.3\%$

(b) (i) $T = \frac{22.48}{10} = 2.25$ s

(ii) Percentage error $= \frac{0.4}{22.48} \times 100\% = 1.78\%$

However, there are events that cannot be repeated, e.g. sprinting competitions. To reduce errors due to reaction time, we can use automatic timing devices. A *light-gate* (Fig 1.4c) is such a device commonly used in the school laboratory.

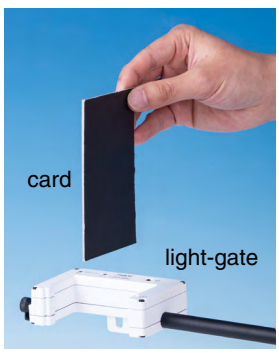


Fig 1.4c *A light-gate.*

light-gate 光閘

A light-gate can measure the time that an object takes to pass through it (Fig 1.4d). The smallest time interval that a light-gate can measure is usually 10^{-3} s.

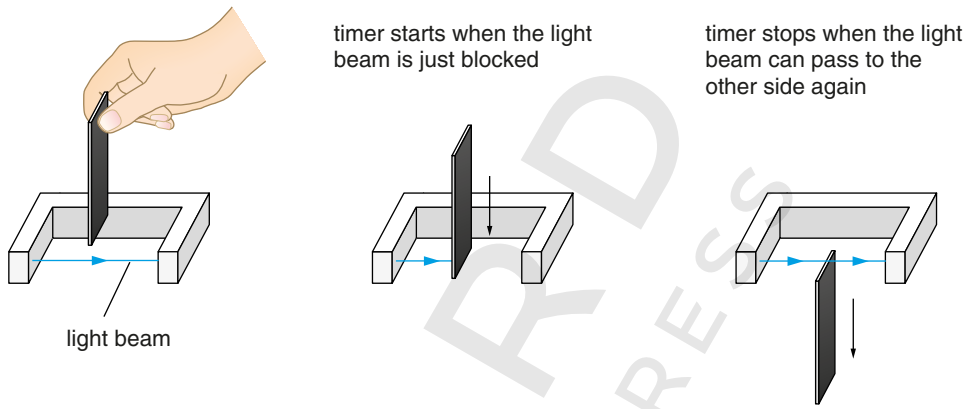


Fig 1.4d *A light-gate automatically measures the time period that the card passes through the device.*

Ts may remind Ss that the choice of a timing device is often made with consideration of the tolerance of percentage error and the practicality of its usage.

2 Error in measurement

Besides the error caused by reaction time, there is another type of error due to the size of gradation of the measuring tool. In Figure 1.4e, the finest gradation of the ruler is 0.1 cm. The measured length of the green line is taken as 3.2 cm. But this is not its actual length. We say that the measured length has an uncertainty of ±0.05 cm.

Uncertainty = finest gradation $\times \frac{1}{2}$

The end of the green line lies between the 3.2 cm and 3.3 cm markings. Since it is closer to the 3.2 cm marking, the measured length reads 3.2 cm, with an uncertainty of ±0.05 cm. With this result, people can know that the actual length L of the green line lies between 3.15 cm and 3.25 cm, i.e. 3.15 cm ≤ L ≤ 3.25 cm.

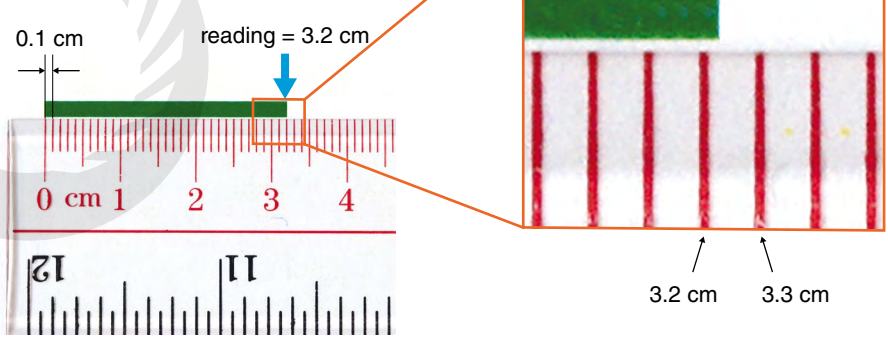


Fig 1.4e *An error in measuring length with ruler.*

Example 6 Percentage error in measurement

Patrick measures the length of an object. His result is 19.1 cm, with an uncertainty of ±0.1 cm. What is the maximum percentage error in Patrick's measurement?

Solution

Maximum error = 0.1 cm

Maximum percentage error = $\frac{0.1}{19.1} \times 100\% = 0.524\%$

Tip
The absolute value of the uncertainty of a measurement gives the maximum error.

Example 7 Reducing percentage error

A line of 9 light dots is produced in an experiment. The dots are evenly spaced by a distance x. John and Mary want to measure the value of x.



(a) John uses a ruler to measure the distance between 2 dots directly. His result is 4 mm, with an uncertainty of ±1 mm. What is the maximum percentage error in this measurement?

(b) Mary uses the same ruler to measure the distance between the first dot and the last dot. Her result is 30 mm, with an uncertainty of ±1 mm.

(i) What is the value of x according to Mary's measurement?

(ii) What is the maximum percentage error in this measurement?

(iii) What is the maximum error in x according to Mary's measurement?

(c) Who obtains a more accurate value of x?

Solution

(a) Maximum percentage error = $\frac{1}{4} \times 100\% = 25\%$

(b) (i) $x = \frac{30}{8} = 3.75$ mm

(ii) Maximum percentage error = $\frac{1}{30} \times 100\% = 3.33\%$

(iii) Maximum error in $x = 3.75 \times 3.33\% = 0.125$ mm

(c) Mary

Tip
The maximum percentage error in x is the same as that in the measurement.

→ Teaching notes and exercises are available on OUP web.

Skill *Experiment*

Error in measurement

All measurements carry errors. To show how significant an error is in a measurement, we should consider the percentage error instead of the size of the error. The smaller the percentage error, the more accurate the measured value.

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Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* A good measurement has no error. (T/F)

2 *True or false:* If the error in the measurement of a length L_1 is larger than that of another length L_2, the error in the measurement of L_1 must be more significant. (T/F)

3 Complete the table below.

	Measured value	Error	Percentage error
(a)	48.9 s	0.1 s	
(b)	12.5 cm		4%

Exercise 1.4

1 The maximum error of the caesium beam atomic clock (Fig a) installed at the Hong Kong Observatory is 10^{-6} s per day. Find the maximum percentage error corresponding to this.



Fig a

A $1.16 \times 10^{-11}\%$ B $1.16 \times 10^{-9}\%$

C $1 \times 10^{-6}\%$ D $1 \times 10^{-4}\%$

★**2** Carol uses a ruler to measure the length of an object. The uncertainty of her measurement is ±1 mm. If the maximum percentage error in her measurement is 5%, what is the length of the object as measured by Carol?

A 1 mm B 2 mm

C 10 mm D 20 mm

★**3** The thickness of 500 pieces of paper is measured to be 5.3 cm. The uncertainty of the measurement is ±0.1 cm. The thickness d of a piece of paper is calculated from the above measurement. What is the maximum error in d?

★**4** Describe the source of error in using a digital stop-watch to measure time due to

(a) the person controlling the stop-watch,

(b) the stop-watch itself.

★**5** A girl is playing on a swing (Fig b). Her father wants to find out the time it takes for her to complete a cycle. He uses a digital stop-watch to measure the time for her to complete 5 cycles and the time recorded is 12.05 s. The uncertainty of the measurement is ±0.4 s. Assume each cycle takes the same amount of time.



Fig b

(a) Estimate the time taken for a cycle.

(b) What is the maximum percentage error in the value obtained in (a)?

(c) Why does the father measure the time for 5 cycles instead of 1 cycle?

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Key concepts 1

1 Time and length

- SI unit of time: second (s)
- SI unit of length: metre (m)

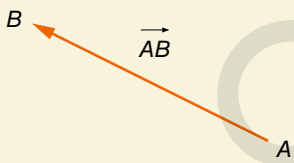
2

Scalar

- Described by magnitude only
- Example: distance

Vector

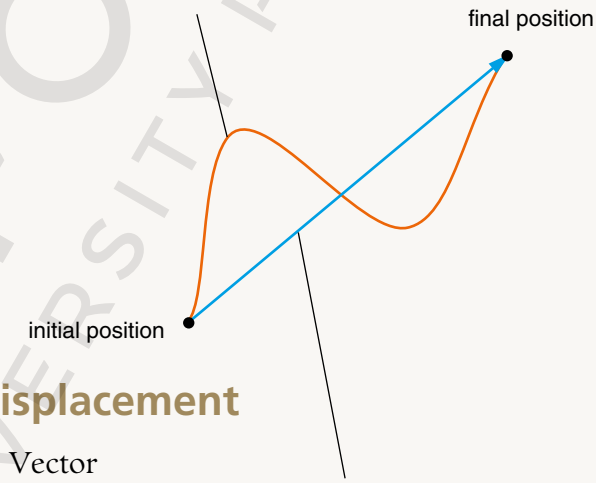
- Require both magnitude and direction
- Can be represented graphically by an arrow



- Example: displacement

Distance

- Scalar
- Length of the path taken



Displacement

- Vector
- Solely depends on the initial and final positions
- Magnitude: length of the straight line join the initial and final positions
- Direction: direction of straight line from the initial to final positions

3

Speed	Velocity
Distance travelled per unit time	Displacement per unit time
Scalar	Vector
Instantaneous speed: speed at an instant	Instantaneous velocity: velocity at an instant, same direction as the direction of motion
Average speed = $\frac{\text{total distance travelled}}{\text{total time of travel}}$	Average velocity = $\frac{\text{total displacement}}{\text{total time of travel}}$, same direction as total displacement
SI unit: m s^{-1}	

4 Acceleration

- Change in velocity per unit time
- Vector
- SI unit: m s^{-2}



Uniform motion

- Motion with a constant velocity (zero acceleration)
- Displacement = velocity × time

5 Error estimation

- Errors exist in all measurements
  - reaction time in using a digital stop-watch (measuring tool)
  - error due to the limitation of the size of graduation of ruler (measuring tool)
- Percentage error = $\frac{\text{error}}{\text{measured value}} \times 100\%$

Key terms

1	acceleration 加速度	p.18
2	average speed 平均速率	p.13
3	average velocity 平均速度	p.15
4	direction 方向	p.7
5	displacement 位移	p.6
6	distance 距離	p.6
7	instantaneous speed 瞬時速率	p.12
8	instantaneous velocity 瞬時速度	p.15
9	magnitude 量值	p.7
10	metre (m) 米	p.3
11	scalar 標量 / 無向量	p.7
12	second (s) 秒	p.2
13	speed 速率	p.11
14	uniform motion 勻速運動	p.20
15	vector 矢量 / 向量	p.7
16	velocity 速度	p.14

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Revision 1

Multiple-choice questions

★1 1.2 A boy walks from X to Z via Y (Fig a). The dotted line shows his path. Which of the following statements is/are correct?

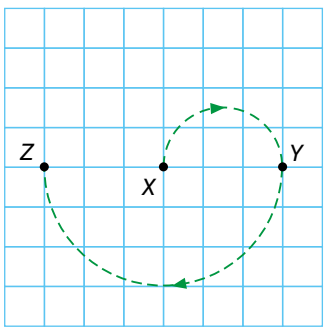


Fig a

(1) The distance travelled from X to Y is equal to the distance travelled from X to Z.

(2) The distance travelled from X to Z is equal to the magnitude of the displacement from X to Z.

(3) The magnitude of the displacement from X to Y is equal to the magnitude of the displacement from X to Z.

A (3) only

B (1) and (2) only

C (1) and (3) only

D (2) and (3) only

★2 1.3 A girl walks with an average speed of 3 km h^{-1} for an hour. Which of the following statements is **incorrect**?

A The magnitude of her total displacement cannot be greater than 3 km.

B Her instantaneous speed can be greater than 3 km h^{-1}.

C The magnitude of her instantaneous velocity can be greater than 3 km h^{-1}.

D The magnitude of her average velocity over the whole trip can be greater than 3 km h^{-1}.

★3 Misc The Monkey King is trapped between the two hands of the Buddha (Fig b). He can only keep flying at a constant speed of 30 m s^{-1} from one of the Buddha's hands to the other along a straight line. He changes his moving direction once he meets the hands. The hands are moving at 10 m s^{-1} towards the Monkey King to catch him. At $t = 0$, the hands are 80 m apart. What is the total distance travelled by the Monkey King from $t = 0$ to the moment the Buddha's hands meet?

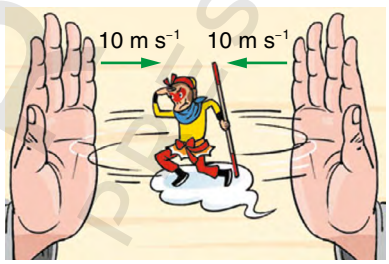


Fig b

A 30 m

B 60 m

C 120 m

D 240 m

★4 1.3 A man walks from P to Q with an average speed v. Then he walks back from Q to P along the same path. What would his average speed in the return trip be so that the average speed of the round trip is $2v$?

A $3v$

B $4v$

C $9v$

D It is impossible.

1.3 **5 HKDSE 2017 Paper 1A Q5**

Which of the following statements about the motion of any two objects is correct?

A The object that takes a shorter time to complete the same path must have greater average speed. (70%)

B The object that travels a greater distance in 1 s must have greater average velocity.

C The object with greater velocity must have greater acceleration.

D If the two objects have the same acceleration, they must be moving in the same direction.

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1.3 **6** **HKDSE 2020 Paper 1A Q4**

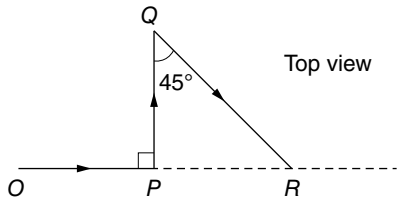


Fig c

A car takes 8 minutes to travel along a path *OPQR* on a horizontal surface as shown. Given that $OP = PQ = 2$ km, find the magnitude of the average velocity of the car in this journey.

A 30 km h^{-1} (64%) B 36 km h^{-1}

C 41 km h^{-1} D 51 km h^{-1}

Conventional questions

*7 1.3 A car in town *X* travels 120 km west to town *Y*, and then travels 50 km south to town *Z*.

(a) Find the magnitude of the total displacement of the car. (2 marks)

(b) If the average speed of the car in this journey is 60 km h^{-1}, find the magnitude of its average velocity. (3 marks)

*8 Misc The table below shows Charles' result in a triathlon (Table a), in which athletes swim 1500 m, cycle 40 km and run 10 km (Fig d).

Swim	Cycle	Run	Total
0:21:28	1:01:53	0:39:47	2:03:08

Table a

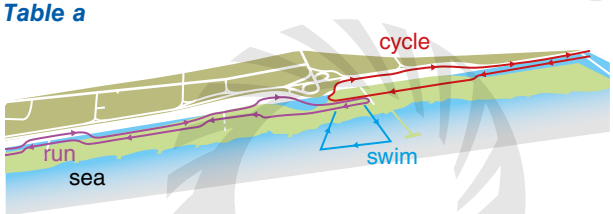


Fig d

(a) What was Charles' average speed over the whole competition? (2 marks)

(b) In which part of the competition (swimming, cycling or running) was his average speed the highest? (1 mark)

(c) In terms of measurement error, do you think that the organizer can use stop-watches to time the results? Explain briefly. (2 marks)

*9 Misc In an experiment estimating the instantaneous speed of a trolley, the trolley takes 0.100 s to move a distance of 4.0 cm.

(a) What instruments should be used in measuring the time and the distance? Give one suggestion for each measurement. (2 marks)

(b) Estimate the instantaneous speed of the trolley according to the measurements. (2 marks)

(c) The uncertainty in the distance measured is ±0.1 cm. The uncertainty in the time measured can be neglected when compared with that of the distance. Find the maximum error in the result in (b). (2 marks)

(d) A student suggests that the result in (b) will be more accurate if he measures the time taken for the trolley to move a much longer distance. Comment on his suggestion. (2 marks)

*10 1.3 Read the following passage about the **quickest path** and answer the questions that follow.

If you travel from one place to another, you may choose the shortest path, i.e. the straight path joining these two places. However, this is not always the quickest path.

For example, a lifeguard at point *X* is going to save a man at point *Y*. Suppose the lifeguard can run at 5 m s^{-1} and swim at 1 m s^{-1}. If he travels along the path shown in Figure e, the time needed will be shorter than travelling along the straight path joining *X* and *Y* (Fig f).

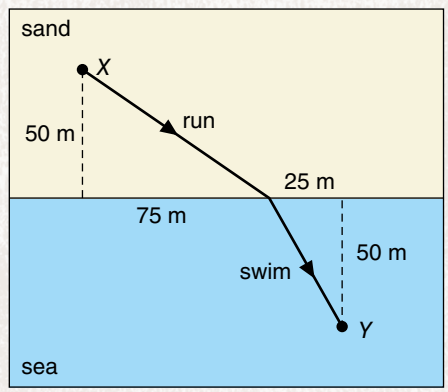


Fig e

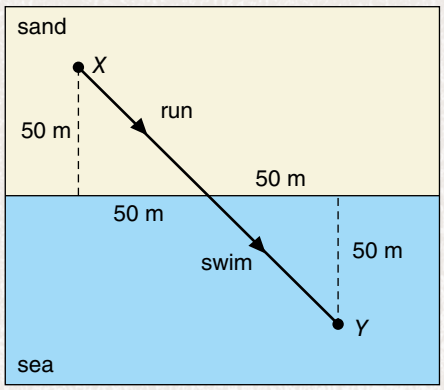


Fig f

(a) Find the total distances travelled by the lifeguard along the paths in Figures e and f. (2 marks)

(b) Find the time needed for the lifeguard to reach the man in these two cases. (2 marks)

(c) Under what circumstance would the straight path joining two points be the quickest path? (1 mark)

★★**11** 1.3 A man walks at velocity v towards the east for time t. Then he walks at velocity $2v$ towards the north for time $2t$ (Fig g). His average speed over the whole journey is 2 m s^{-1}.

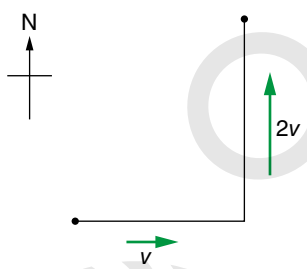


Fig g

(a) Find the direction of his average velocity over the whole journey. (2 marks)

(b) Find the magnitude of his average velocity over the whole journey. (3 marks)

Refer p.16

★★**12** Misc A toy car is released from rest at X on an inclined plane (Fig h). It travels along a straight line to Y via Z.

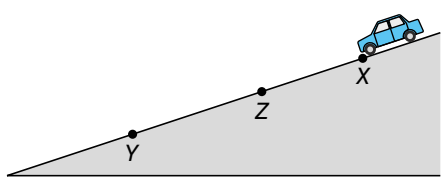


Fig h

You are given a measuring tape, a digital stop-watch, a light-gate, a piece of card and some adhesive tape.

(a) Describe an experiment to measure the average speed of the toy car travelling from X to Y. (3 marks)

(b) Describe an experiment to measure the instantaneous speed of the toy car at Z. (5 marks)

Refer p.26

☆ Physics A Unit G481/01

1.3 **13** **OCR GCE AS Jan 2013 Q2**

(a) Speed is a scalar quantity and velocity is a vector quantity. State one difference and one similarity between speed and velocity. (2 marks)

(b) Figure i shows a toy locomotive on a circular track.

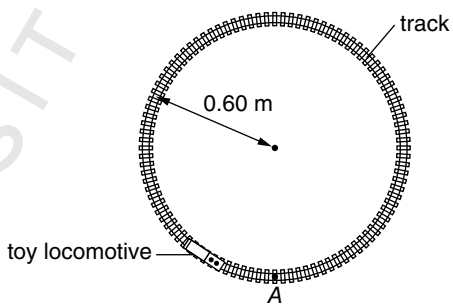


Fig i

The locomotive travels at constant speed round the track in a clockwise direction. It takes 12 s to travel completely round the track. At time $t = 0$, the locomotive is at point A.

(i) Calculate the speed of the locomotive. (2 marks)

(ii) Calculate the magnitude of the displacement s of the locomotive from point A after it has travelled one quarter of the way round the track. (2 marks)

(iii) Explain why the average velocity of the locomotive is zero after a time of 12 s. (1 mark)

(iv) Explain why the velocity of the locomotive changes even though its speed is constant. (1 mark)

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Know more

Distance posts (p.7)

In Hong Kong, distance posts are installed every 500 m along some trails to help hikers identify their positions. If you walk from the starting point of the MacLehose Trail to the post M158 along the trail, you have walked a distance of 79 000 m (= 500 × 158).

In the figure, there is another number KK012805 marked on the post. It is the grid reference on the map. It shows your exact position. The vector pointing from the starting point to this position is your displacement.



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